of the Moon or Venus or Earth. Deeper than 300 kilometers, the pressures should transform carbon into diamond. We know that Mars has been geologically active over its history. Material from great depth will occasionally be extruded up to the surface, and not just in the great volcanos. So there does seem to be a case for diamonds on other worlds—on Mars, and not the Moon. In what quantities, of what quality and size, and in which locales we do not yet know.

The return to Earth of a spacecraft stuffed with gorgeous multicarat diamonds would doubtless depress prices (as well as the shareholders of the de Beers and General Electric corporations). But because of the ornamental and industrial applications of diamonds, perhaps there is a lower limit below which prices will not go. Conceivably, the affected industries might find cause to promote the early exploration of Mars.

The idea that Martian diamonds will pay for exploring Mars is at best a very long shot, but it's an example of how rare and valuable substances may be discoverable on other worlds. It would be foolish, though, to count on such contingencies. If we seek to justify missions to other worlds, we'll have to find other reasons.

BEYOND DISCUSSIONS OF PROFITS and costs, even reduced costs, we must also describe benefits, if they exist. Advocates of human missions to Mars must address whether, in the long term, missions up there are likely to mitigate any of the problems down here. Consider now the standard set of justifications and see if you find them valid, invalid, or indeterminate:

Human missions to Mars would spectacularly improve our knowledge of the planet, including the search for present and past life. The program is likely to clarify our understanding of the environment of our own planet, as robotic missions have already begun to do. The history of our civilization shows that the pursuit of basic knowledge is the way the most significant practical advances come about. Opinion polls suggest that the most popular reason for "exploring space" is "increased knowledge." But are humans in